

Solving equations using evidence and zero.

So, we have (as an example)

$$x^4 + 3x^3 + 5x^2 + x = 2$$

Subtract 2 by both sides

$$x^4 + 3x^3 + 5x^2 + x - 2 = 0$$

Put x in evidence.

$$x(x^3 + 3x^2 + 5x + 1 - \frac{2}{x}) = 0$$

$$x^3 + 3x^2 + 5x + 1 - \frac{2}{x} = \frac{0}{x}$$

So, dividing zero, by x, right side

$$x^3 + 3x^2 + 5x + 1 - \frac{2}{x} = 0$$

putting x in evidence again

$$x(x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2}) = 0$$

$$x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2} = \frac{0}{x}$$

dividing zero by x, on right side.

Equation 1

$$x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2} = 0$$

End Equation 1

In this, case , we have a parabola formula (On equation 1)

Also,

$x^4 + 3x^3 + 5x^2 + x = 2$ is a parabola formula because of equation 1

To consider

On a middle of a calculation, you can convert some expressions, into others, making it easier to calculate.

For example

$$A^{x-1} * C - A^x = B^y \quad (1)$$

$$A^x \left(\frac{1}{A} * C - 1 \right) = B^y \quad (2)$$

$$A^x = \frac{B^y}{\frac{C}{A} - 1} \quad (3)$$

$$A^x = \frac{B^y}{\frac{C-1}{A}} \quad (4)$$

$$A^x * A = \frac{B^y}{C-1} \quad (5)$$

$$A^x * A * (C-1) = B^y \quad (6)$$

Now substitute , on (6), the B^y of (2). We get

$$\text{left side is (6)} \qquad \text{right side is (2)}$$

$$A^x * A * (C-1) = A^{\left(\frac{1}{A} * C - 1\right)}$$

Also

So, we can solve the equation by solving 1,2 and 6...no need extra arithmetic. (steps 3,4 and 5).

Inequations

So to solve inequations

$$3 < N < 7$$

reduce left side to zero

$$3-3 < N < 7-3$$

$$0 < N < 4$$

you can now multiply whatever you want (since left side will always be zero)

Using my calculus of beals conjecture.. As you can

$$\text{Postulated}$$

$$C = B^y$$

$$A^x > B^y$$

We have the equations..

Other approach

Consider start at solution, (since its easier to verify the problem), and brute force it. Just substitute equations, by a number, should test 0-0xffffffff variables. (which is about 20 to 15 minutes)

Say start at $x=2$

We have equation

$$x^4 + 3x^3 + 5x^2 + x = 2$$

$$2*2*2*2 + (3*2)^3 + 5(2) + 2 = 2$$

Doesnt hold.

X=3....and so on...